

Calculus and Analytical Geometry

Intervals, Inequalities, and Absolute Values

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Objectives

After preparing this topic, you should be able

1. To represent and interpret intervals and inequalities on the real number line.
2. To understand absolute value as distance and solve absolute-value equations and inequalities.
3. To translate distance statements into intervals and use them to understand the language of limits.

The Why Section

Objective 1: Why Study Intervals and Inequalities?

- Before studying **limits**, we need a precise language for describing **where a variable can be and how close it can get to a particular number**.
- Intervals and inequalities provide this language.
- We study these concepts because they allow us to:
 - **Describe the location of numbers on the real number line**
 - Inequalities and intervals help us clearly identify regions such as

$(x < 4)$, $(x > 2)$, or $(2 < x < 5)$.

- This is essential for describing the values of (x) near a point.
- **Express whether endpoints are included or excluded**
 - The distinction between $(<)$ and $(\leq)$, and between parentheses $((,))$ and brackets $([,])$, becomes crucial when discussing **open intervals around a number**.
 - Limits frequently consider values *near* (c) without necessarily allowing $(x = c)$.
- **Describe the domain and conditions under which functions are studied**
 - Many functions are defined only on certain intervals or under certain inequalities.
 - Being able to translate between inequalities, interval notation, and the number line prepares us to describe **domains, neighborhoods, and the values of (x) used in limits**.

Connection to Limits

- The central idea we will soon need is:
 - **A limit studies what happens to $(f(x))$ when (x) gets close to a particular number (c) .**
- To say that (x) is **close to** (c) , we need intervals and inequalities such as the following two statement.

$$c - \delta < x < c + \delta$$

$$x \in (c - \delta, c + \delta).$$

- Thus, **intervals and inequalities are the mathematical language that allows us to describe “close to” precisely**—one of the fundamental ideas behind limits.

Objective 2: Why Study Absolute Value as

Distance?

- After learning how **inequalities and intervals describe the position of numbers**, we now need a way to describe **how far one number is from another**.
- Absolute value provides exactly this idea and creates a direct bridge to limits.
- We study absolute value because we need to:
 - **Measure the distance between two numbers**
 - The expression $|x - c|$ represents the distance between (x) and (c) .
 - This gives us a precise mathematical meaning for phrases such as “**close to (c)** ” or “**within a certain distance of (c)** ”.
 - **Describe numbers near a particular point**
 - Expressions such as $|x - c| < a$ tell us that (x) lies within (a) units of (c) and mathematically
$$c - a < x < c + a.$$
 - This is exactly the language we will use when discussing values of (x) **approaching a number**.
 - **Build the foundation for the precise definition of a limit**
 - The concept of a limit depends on controlling the distance between (x) and (c) , and later between $(f(x))$ and (L) .
 - The expressions $|x - c| < \delta$, $|f(x) - L| < \varepsilon$ are at the heart of the formal definition of a limit.

Connection to Limits

- The key idea we are preparing for is:
 - **A limit asks what happens to $(f(x))$ when (x) gets arbitrarily close to (c) .**

- The phrase “ (x) **gets close to** (c) ” can be expressed precisely as

$$|x - c| < \delta$$

- Therefore, understanding **absolute value as distance** is not merely an algebraic technique—it gives us the mathematical language needed to make the intuitive idea of **closeness** precise in calculus.

Objective 3: Why Study “From Distance to the Language of Limits”?

- We study distance as the language of limits because:
 - **Make the idea of “closeness” precise**
 - In everyday language, “close to (c) ” is vague.
 - The expression $|x - c| < \delta$ gives a precise mathematical meaning: (x) is less than (δ) units away from (c) .
 - This allows us to replace informal descriptions with exact conditions.
 - **Understand what “approaching” means in a limit**
 - When we write $x \rightarrow c$, (x) gets closer and closer to (c) , but (x) does not have to equal (c) .
 - The condition $0 < |x - c| < \delta$ captures this idea precisely and introduces the concept of a **deleted neighborhood**.
 - **Build the foundation for the formal definition of a limit**
 - A limit connects two kinds of closeness:

$$|x - c| \rightarrow (\text{input distance})$$

$$|f(x) - L| \rightarrow (\text{output distance})$$

- Understanding these distances prepares us to understand the precise statement that **making (x) sufficiently close to (c) makes $(f(x))$ sufficiently close to (L)** .

Connection to Limits

- The entire purpose of this section is to prepare us for the language of the limit:

$$0 < |x - c| < \delta \implies |f(x) - L| < \varepsilon$$

- In words, “If (x) is sufficiently close to (c), but not equal to (c), then (f(x)) is sufficiently close to (L).”
- Thus, this objective forms the **bridge between the elementary concepts of intervals, inequalities, and distance and the formal mathematical definition of a limit.**

Objective 1: Intervals and Inequalities

The Real Number Line

- The real number line gives us a way to represent every real number as a point.

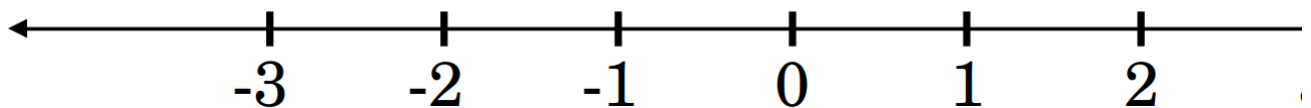


Figure 1. The real number line.

- Moving to the right means numbers become larger.
- Moving to the left means numbers become smaller.
- For example, $-2 < 3$ because -2 lies to the left of 3 .

Inequalities

- An inequality compares two quantities.
- The four basic symbols are:

1. $<$ $\implies$ Less than symbol

2. $>$ $\implies$ Greater than symbol

3. $\leq$ $\implies$ Less than or equal to symbol

4. $\geq$ $\implies$ Greater than or equal to symbol

- For example, $x < 4$ means x is somewhere to the left of 4.
- Similarly, $x > 4$ means x is somewhere to the right of 4.

Open and Closed Endpoints

- Consider $x < 4$.
- The number 4 is not included.
- We represent this using an open endpoint $\circ$.
- But, $x \leq 4$ includes 4, so we use a filled endpoint $\bullet$.

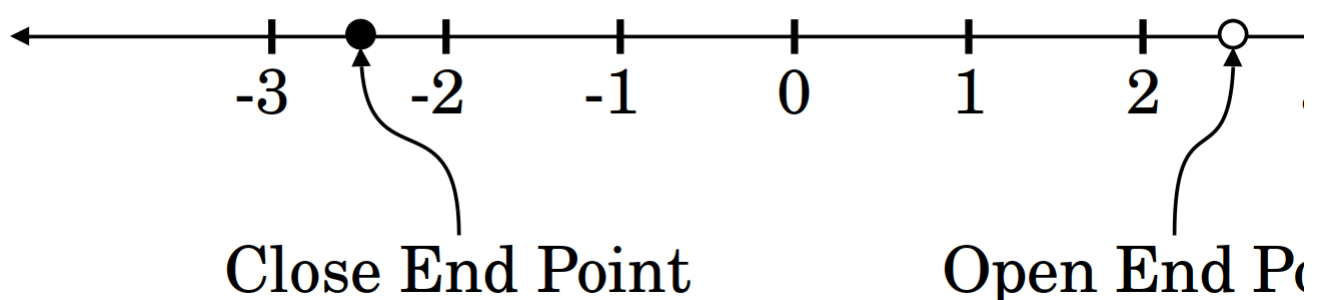


Figure 2. Open and close end points on number line.

- This distinction will later matter when we discuss open intervals around the point c .

Intervals

- An interval is a set of real numbers between two endpoints.
- For example, $2 < x < 5$ means all numbers between 2 and 5.
- In interval notation $(2, 5)$, the parentheses mean that 2 and 5 are not included.

2 and 5 are not included

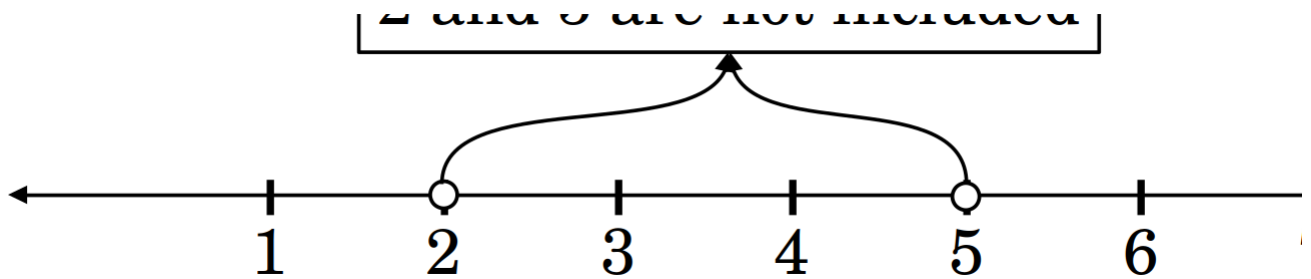


Figure 3. Representation of $(2, 5)$ on real number line.

Important Types of Intervals

- Open interval $a < x < b$ becomes (a, b) .
- Closed interval $a \leq x \leq b$ becomes $[a, b]$.
- Left-closed, right-open $a \leq x < b$ becomes $[a, b)$.
- Left-open, right-closed $a < x \leq b$ becomes $(a, b]$.

Intervals Extending to Infinity

- For $x > 3$, we write $(3, \infty)$.
- For $x \leq 3$, we write $(-\infty, 3]$.
- Remember that ∞ is not a real number, so we always use a parenthesis with infinity.

Example 1

Write $-2 < x < 4$ in interval notation.

Solution

The endpoints are both excluded. Therefore, $x \in (-2, 4)$



Example 2

Write $[-3, 5)$ as an inequality.

Solution

The square bracket at -3 means included, so

$$x \geq -3$$

The parenthesis at 5 means excluded, so

$$x < 5$$

Therefore,

$$\boxed{-3 \leq x < 5}$$



Example 3

Solve

$$3x - 2 < 10$$

Solution

Adding 2 to both sides, we get

$$3x < 12$$

Dividing both sides of above inequality by 3 , we get

$$x < 4$$

Therefore,

$$\boxed{x \in (-\infty, 4)}$$



Example 4

Solve

$$-2x + 3 < 7$$

Solution

Subtracting 3 from both sides of the given inequality, we get

$$-2x < 4$$

Dividing both sides of above inequality by -2 , we get

$$x > -2$$

Therefore,

$$x \in (-2, \infty)$$

Important Note

Multiplication or division by a negative number reverses the inequality.



Example 5

Solve

$$2 < 3x - 1 < 8$$

Solution

Adding 1 to all three parts of the above inequality, we get

$$3 < 3x < 9$$

Dividing all parts of above inequality by 3, we get

$$1 < x < 3$$

Therefore,

$$x \in (1, 3)$$



Quick Check

Question No. 1

Express the inequality

$$-4 \leq x < 3$$

in interval notation.

Question No. 2

Solve

$$-3x + 6 \geq 12$$

and express your answer in interval notation.



Objective 2: Absolute Value as Distance

- Now we move from order to distance.
- This is where the material becomes directly connected to limits.

What Is Absolute Value?

- The **absolute value of a number** is its **distance from zero**.
- For example, $|5| = 5$ because 5 is 5 units from zero.
- Also, $|-5| = 5$ because -5 is also 5 units from zero.

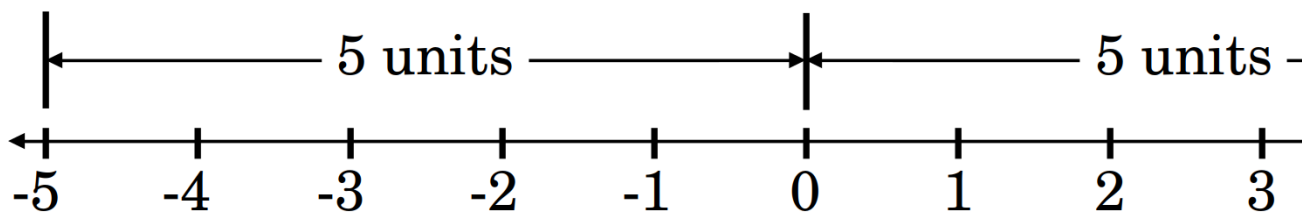


Figure 4. Absolute value as distance from 0.

Therefore,

$$|x| \geq 0$$

Absolute Value as Distance Between Two Numbers

- The distance between x and c is $|x - c|$.
- For example, the distance between $x = 3$ and $c = 7$ is

$$|3 - 7| = |-4| = 4 = |7 - 3|$$

The distance between $x = 9$ and $c = 7$ is

$$|9 - 7| = 2 = |7 - 9|$$

Thus,

$$|x - c| = \text{distance from } x \text{ to } c$$

- This interpretation is much more important for limits than simply memorizing the definition of absolute value.

Absolute-Value Equations

- Consider $|x| = 3$.
- Which numbers are exactly 3 units from zero?
- The numbers $x = 3$ and $x = -3$ are exactly 3 units from zero.
- Therefore,

$$|x| = 3 \iff x = \pm 3$$

Example 6

Solve

$$|x - 4| = 2$$

Solution

The equality $|x - 4| = 2$ means x is exactly 2 units from 4.

Therefore,

$$x - 4 = 2 \quad \text{or} \quad x - 4 = -2$$

Thus,

$$x = 6 \quad \text{or} \quad x = 2$$

Hence,

$$x = 2, 6$$



Absolute-Value Inequalities

- Now consider

$$|x| < 3$$

- This means the distance of x from zero is less than 3.
- Therefore x must lie between -3 and 3 , i.e.,

$$-3 < x < 3$$

- Thus,

$$|x| < 3 \iff -3 < x < 3$$

Fundamental Rule

- For $a > 0$

$$|x| < a \iff -a < x < a$$

- More generally,

$$|x - c| < a \iff c - a < x < c + a$$

- This formula is extremely important for limits.

Example 7

Solve

$$|x - 5| < 2$$

Solution

Write the equivalent compound inequality

$$-2 < x - 5 < 2$$

$$3 < x < 7$$

Therefore,

$$x \in (3, 7)$$

Logic

$|x - 5| < 2$ means x is within 2 units of 5. So the interval is $(5 - 2, 5 + 2) = (3, 7)$.



Absolute Value Greater Than a Number

- Consider $|x| > 3$.
- This means x is more than 3 units away from zero.
- Therefore, x must be outside the interval $[-3, 3]$.
- It means $x < -3$ or $x > 3$.
- Hence,

$$|x| > 3 \iff x < -3 \text{ or } x > 3$$

- This is an important contrast and the following key helps to remember it.

$$|x| < a \rightarrow \text{Between} \qquad |x| > a \rightarrow \text{Outside}$$

Example 8

Solve

$$|x - 2| > 4$$

Solution

We have two possibilities

$$x - 2 < -4 \quad \text{or} \quad x - 2 > 4$$

Therefore,

$$x < -2 \quad \text{or} \quad x > 6$$

Hence,

$$x \in (-\infty, -2) \cup (6, \infty)$$



Example 9

Consider

$$|2x - 8| < 2$$

Solution

We write

$$-2 < 2x - 8 < 2$$

Adding 8 to all parts of above expression

$$6 < 2x < 10$$

Dividing all parts of above expression by 2

$$3 < x < 5$$

Therefore,

$$x \in (3, 5)$$

Or, equivalently,

$$|x - 4| < 1$$

This last form is particularly important because it says, “ x is within 1 unit of 4”. That is precisely the idea needed for limits.



Quick Check

Question 1

Solve

$$|x - 3| < 4$$

Question 2

Solve

$$|2x + 4| \leq 6$$



Objective 3: From Distance to the Language of Limits

- Now we connect everything.
- This is the most important objective of the lecture.

What Does “Close to (c)” Mean?

- Suppose we say, “ x is close to c ”.
- Mathematically, we need to specify how close.
- Suppose we choose a positive number δ , such that

$$|x - c| < \delta$$

- Then x is less than δ units away from c .

- Using our absolute-value rule

$$-\delta < x - c < \delta$$

Add c to all parts.

$$c - \delta < x < c + \delta$$

Therefore,

$$|x - c| < \delta \iff c - \delta < x < c + \delta$$

And in interval notation, we can write

$$|x - c| < \delta \iff x \in (c - \delta, c + \delta)$$

This Is a Neighborhood Around c

- The interval $(c - \delta, c + \delta)$ is centered at c .
- Its radius is δ .
- For example, let

$$c = 4, \quad \delta = 1$$

- Then $(c - \delta, c + \delta)$ becomes $(3, 5)$.
- Thus,

$$|x - 4| < 1$$

- The above inequality means exactly the same thing as

$$3 < x < 5$$

Example 10

Rewrite

$$|x - 6| < 0.5$$

as an interval.

Solution

Using $|x - c| < \delta \iff c - \delta < x < c + \delta$, we get

$$6 - 0.5 < x < 6 + 0.5$$

Therefore,

$$5.5 < x < 6.5$$

Or,

$$x \in (5.5, 6.5)$$



What Does “Approaching c ” Mean?

- When we write $x \rightarrow c$, we mean that x gets closer and closer to c .
- But in a limit, x is not required to equal c .
- Therefore we use

$$0 < |x - c| < \delta$$

- The condition $|x - c| < \delta$ puts x inside the neighborhood.
- The condition $0 < |x - c|$ removes the point $x = c$.
- Thus, $0 < |x - c| < \delta$ means, x is within δ units of c , but $x \neq c$.

Deleted Neighborhood

- The interval $(c - \delta, c + \delta)$ contains c .
- But $0 < |x - c| < \delta$ excludes c .
- Therefore, $(c - \delta, c) \cup (c, c + \delta)$ is the corresponding deleted neighborhood.
- This distinction is important because the definition of a limit specifically

uses the following expression

$$0 < |x - c| < \delta$$

- The PDF emphasizes that the value of $f(c)$ itself does not influence the existence of the limit.

One-Sided Neighborhoods

- Sometimes we approach c only from one side.
- From the right

$$x > c$$

- Within δ units

$$c < x < c + \delta$$

- From the left

$$x < c$$

- Within δ units

$$c - \delta < x < c$$

- These are the interval structures that appear later in the PDF's definitions of right-hand and left-hand limits.

Example 11

Let

$$c = 3, \quad \delta = 0.2$$

Describe the points within 0.2 units of 3, from the right.

Solution

We need

$$3 < x < 3.2$$

Therefore,

$$x \in (3, 3.2)$$



Example 12

Let

$$c = 3, \quad \delta = 0.2$$

Describe the points within 0.2 units of 3, from the left.

Solution

We need

$$2.8 < x < 3$$

Therefore,

$$x \in (2.8, 3)$$



From Output Distance to Input Distance

- Now we reach the exact idea that motivates the precise definition of a limit.
- Suppose

$$f(x) = 2x - 1$$

- We want $f(x)$ to be within 2 units of 7.
- We write $|f(x) - 7| < 2$.
- Substitution of $f(x)$ in the above inequality gives

$$|2x - 1 - 7| < 2$$

- Thus,

$$|2x - 8| < 2$$

- From Objective 2, $3 < x < 5$.

- So we have discovered

$$|f(x) - 7| < 2 \quad \text{whenever} \quad 3 < x < 5$$

- Since $3 < x < 5$ is equivalent to $|x - 4| < 1$, so we can write

$$|f(x) - 7| < 2 \quad \text{whenever} \quad |x - 4| < 1$$

- This reasoning will be used in solving limits.

The Two Types of Closeness

- This gives us two different distances.

Input distance

- The distance $|x - c|$ measures how far x is from c .

Output distance

- The distance $|f(x) - L|$ measures how far $f(x)$ is from L .
- For a limit, $x \rightarrow c$ means we control $|x - c|$.
- And we want this to force $|f(x) - L|$ to become small.

The Language We Are Preparing For

- The precise definition of a limit says

$$|f(x) - L| < \epsilon \quad \text{whenever} \quad 0 < |x - c| < \delta$$

- We are not proving this definition yet.

- We are simply understanding its language.
- The expression $0 < |x - c| < \delta$ means x is close to c , but $x \neq c$.
- The expression $|f(x) - L| < \epsilon$ means $f(x)$ is close to L .
- The PDF introduces exactly this definition after motivating the need to replace the vague phrase “gets arbitrarily close” with precise conditions.

Example 13

Suppose

$$|x - 5| < 0.1$$

What interval contains x ?

Solution

Using $|x - c| < \delta \iff c - \delta < x < c + \delta$, we have $5 - 0.1 < x < 5 + 0.1$.

Therefore,

$$4.9 < x < 5.1$$

Or,

$$x \in (4.9, 5.1)$$



Example 14

Suppose

$$0 < |x - 5| < 0.1$$

Find the interval in which x exists.

Solution

Then

$$4.9 < x < 5.1$$

but

$$x \neq 5$$

Therefore,

$$x \in (4.9, 5) \cup (5, 5.1)$$

Important Note

This is exactly the type of input restriction used in the precise definition of a limit.



Example 15

Suppose x must remain inside $(2, 10)$ and we want a symmetric interval around 5

$$(5 - \delta, 5 + \delta)$$

How large can δ be?

Solution

Distance from 5 to the left endpoint

$$5 - 2 = 3$$

Distance from 5 to the right endpoint

$$10 - 5 = 5$$

The nearer endpoint is 2.

Therefore,

$$\delta = 3$$

This is the largest possible symmetric radius.

Then

$$(5 - 3, 5 + 3) = (2, 8)$$

Thus,

$$(2, 8) \subset (2, 10)$$

Important Note

This idea will also be used in limits.



Quick Check

Question 1

Rewrite

$$|x - 4| < 0.2$$

as an interval.

Question 2

If

$$0 < |x - 3| < 0.5$$

Describe the corresponding interval(s) for x .



Summary

- Today we built the mathematical language needed for limits.

- We learned how to describe regions of the real number line

$$a < x < b \iff x \in (a, b)$$

- We also learned how to solve inequalities and remember, “multiplying or dividing by a negative reverses the inequality”.
- The most important interpretation is

$$|x - c| = \text{distance between } x \text{ and } c$$

- Therefore, $|x - c| < a$ means that x lies within a units of c .
- Equivalently,

$$|x - c| < a \iff c - a < x < c + a$$

- We learned the crucial translation

$$|x - c| < \delta \iff c - \delta < x < c + \delta$$

- Therefore

$$|x - c| < \delta \iff x \in (c - \delta, c + \delta)$$

- If $x = c$ is excluded $0 < |x - c| < \delta$ means

$$x \in (c - \delta, c) \cup (c, c + \delta)$$

- Finally, we can now understand the basic language behind the precise definition of a limit

$$0 < |x - c| < \delta \implies |f(x) - L| < \epsilon$$

- In words,
- If x is sufficiently close to c , but not equal to c , then $f(x)$ is as close to L as we want.
- That is why intervals, inequalities, and absolute values are the essential prerequisites for understanding the precise definition of a limit.

Exercises

Important Note: The Lucky Day

This is your lucky day, we don't have exercises. But, you need to solve all the examples with their logical understanding. If you understand limits, you will be able to understand continuity. And, if you understand continuity, you will be able to understand logical understanding of the derivatives.